

**The University of Texas at Austin**  
**IN-CLASS WORK 6**

*M339D Intro to Financial Math*

September 21, 2026

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**HEDGING WITH EUROPEAN CALL OPTIONS**

**PROBLEM 6.1:** Sample FM(DM) #13.

*Suppose that you short one share of a stock index for 50, and that you also buy a 60 —strike European call option that expires in 2 years for 10. Assume the effective annual interest rate is 3%. If the stock index increases to 75 after 2 years, what is the profit on your combined position?*

**PROBLEM 6.2:** *The “Very tasty goat cheese Co” sells artisan goat cheese at \$10 per oz. They need to buy 200 gallons of goat milk in six months to make 200 oz of their specialty fall-equinox cheese. Non-goat milk aggregate costs total \$500. They decide to buy six-month, \$5-strike call options on gallons of goat milk for 0.50 per call option.*

*The continuously compounded, risk-free interest rate equals 0.04.*

*In six months, the price of goat milk equals \$6 per gallon. What is the profit of the company’s hedged position?*

**PROBLEM 6.3:** *The primary ingredient for a certain jeweler is gold which she intends to buy in exactly one year. She considers all of her other production-related expenses to be negligible.*

*The jeweler uses exactly one ounce of gold to produce every one of her pieces, and will be able to sell every piece for \$1,000.*

*The jeweler models the market price of gold in one year as follows:*

| <i>Gold price in one year</i> | <i>Probability</i> |
|-------------------------------|--------------------|
| <i>750 per ounce</i>          | <i>0.2</i>         |
| <i>850 per ounce</i>          | <i>0.5</i>         |
| <i>950 per ounce</i>          | <i>0.3</i>         |

*The jeweler hedges the price of gold by buying a 1 — year call option with an exercise price of \$900 per ounce. The option costs \$10 per ounce now.*

*The continuously compounded risk-free interest rate is 5%.*

*Calculate the expected profit of the **hedged** portfolio per piece of jewelry produced.*

**PROBLEM 6.4:** Suppose that you own one share of a stock index, currently priced at \$50, and that you write a \$60-strike European call option on it that expires in 2 years for \$10. Assume the effective annual interest rate is 3%.

You model the stock index price in 2 years as follows:

| Stock index price in 2 years | Probability |
|------------------------------|-------------|
| 40                           | 0.25        |
| 65                           | 0.50        |
| 90                           | 0.25        |

Calculate the expected **payoff** of your **covered call** position.

**PROBLEM 6.5:** *A silver dealer owns exactly one ounce of silver, which — being non-perishable — she could sell at any time, but intends to sell in exactly one year. She considers all other costs negligible.*

*She models the market price of silver in one year as follows:*

| <i>Silver price in one year</i> | <i>Probability</i> |
|---------------------------------|--------------------|
| <i>18 per ounce</i>             | <i>0.5</i>         |
| <i>22 per ounce</i>             | <i>0.3</i>         |
| <i>26 per ounce</i>             | <i>0.2</i>         |

*To generate extra income, the dealer writes a 1-year covered call on her silver with strike price \$20 per ounce, and receives \$1.20 per ounce now for selling the option.*

*The continuously compounded, risk-free interest rate is 4%.*

*Calculate the expected profit of her **covered call** portfolio per ounce.*

**REMARK 6.6:** Why would the dealer write this call rather than simply hold the silver unhedged? The median outcome in her distribution is \$18, below the \$20 strike, so the price she thinks **most likely** is one where the call expires worthless and she keeps the full premium — even though the mean, \$20.80, sits just above the strike because of the less-likely but higher \$26 outcome pulling it up. Since she already owns the ounce and is selling it in a year regardless, writing the call is not a directional bet; it monetizes her view that a rise much above \$20 is the less probable outcome. The \$1.20 premium lowers her effective cost basis and cushions a price decline, at the cost of capping her proceeds if silver does rally past \$20 — the usual trade-off of a covered call: income generation and partial downside protection, paid for by giving up the upper tail.